

DATA3: Single-Salt NF270 Diafiltration

Concentration-dependent solute permeability & the Donnan–dielectric model

Collaborator update • June 24, 2026

Keshav Kasturi Rangan • University of Notre Dame

Feedback from Friday — addressed

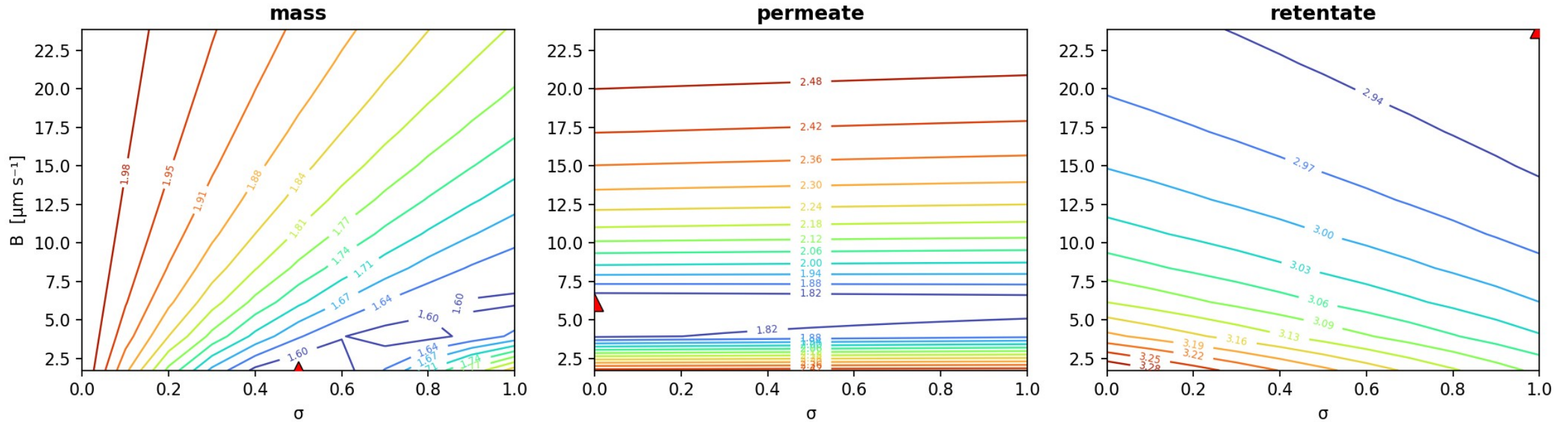
- **Time-series correction:** permeate ICP now anchored at vial close (DATA2 convention); mass, retentate, and permeate share one time axis — the invented tube-transit placement is retired.
- **Identifiability contours:** $\sigma \times B$ log-WSSE surfaces generated per response (mass / permeate / retentate) at the identified L_p .
- **Color coding:** fit/data plots follow the DATA2 convention (retentate vs permeate vs model).
- **Scope:** all changes are DATA3 / NF270-only; DATA1, DATA2, and the legacy MATLAB are untouched.

Topics I need help with

- **Prioritization:** which question below is the right one to chase first?
- **Donnan-dielectric equation:** the partition form, its assumptions, and the derivation of $B(c)$.
- **Interfacial concentration:** what range of membrane-interface concentration do the experiments actually probe?
- **Péclet framing:** the Péclet number and the rejection law $R(\sigma, Pe)$ — is this the right lens?
- **Transport components:** dielectric exclusion, cross-coefficients D_{ij} , $B(c)$, and activity coefficients — which to include?
- **Identifiability:** given σ is unidentified single-experiment, how far can $B(c)$ claims go?

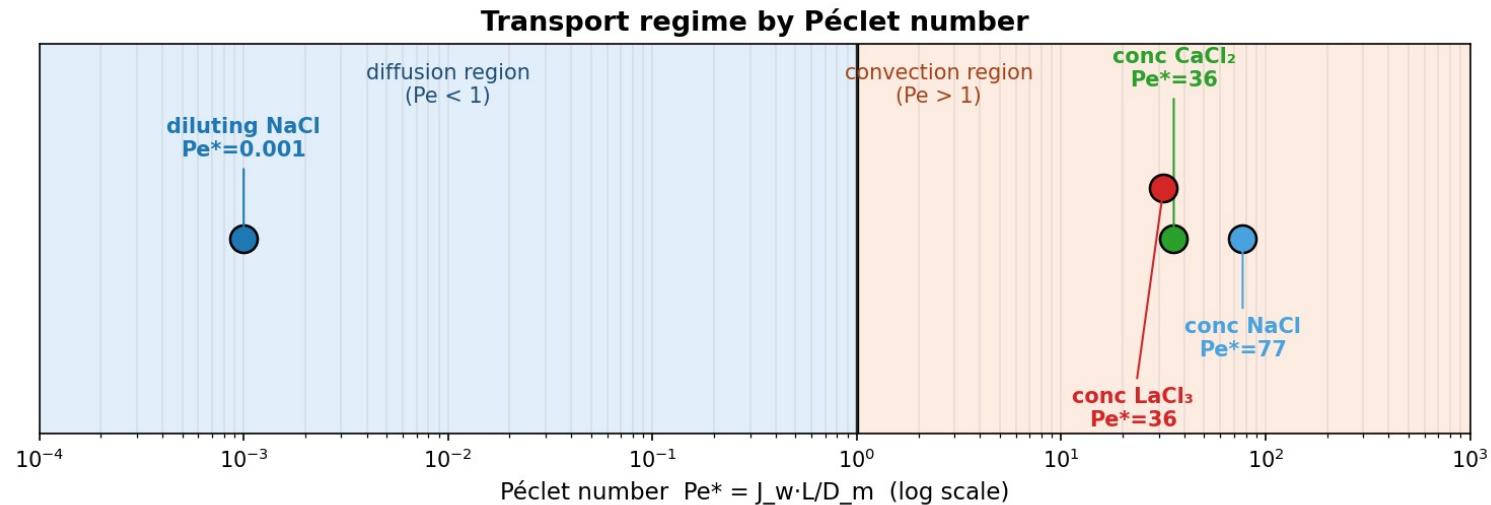
Single experiments determine B, not σ

concentrating CaCl_2 (MC2.05.07.24) — $\sigma \times B$ identifiability, hydraulic permeability fixed at its identified optimum $L_p = 7.54 \text{ L m}^{-2} \text{ h}^{-1} \text{ bar}^{-1}$



- Each response's log-WSSE has a narrow valley in B but is nearly flat in σ (representative concentrating CaCl_2 ; L_p fixed at its identified optimum).
- σ rails to its bound while B stays well-determined — the two are correlated, so one experiment cannot recover σ .
- **Consequence:** “is B concentration-dependent?” must be posed carefully — we lean on mechanism, not fit alone.

The same membrane spans diffusion- and convection-limited regimes



$$Pe = \text{convection} / \text{diffusion} = J_w \cdot L / D_m$$

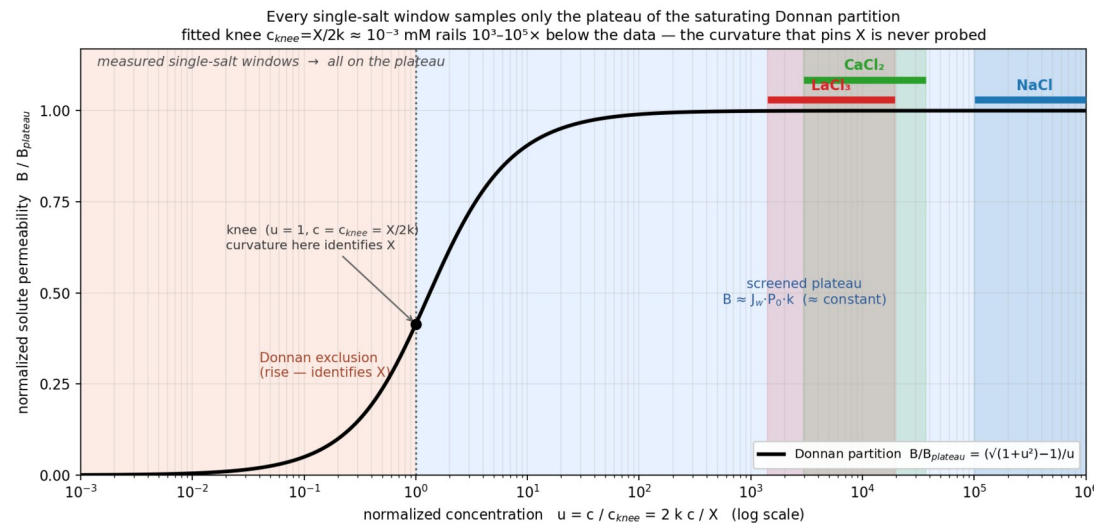
$$R = \sigma(1 - e^{-Pe}) / (1 - \sigma e^{-Pe})$$

where Pe — Péclet number; J_w — water (volume) flux through the membrane; L — membrane thickness; D_m — solute diffusivity in the membrane; R — salt rejection; σ — reflection coefficient (0 = no rejection, 1 = complete).

- **Diluting NaCl is diffusion-limited ($Pe^* \approx 10^{-3}$); concentrating NaCl / CaCl_2 / LaCl_3 are convection-dominated ($Pe^* \approx 77 / 36 / 36$).**
- **Mechanistic inference, not a direct demonstration:** Pe and B come from the same fitted model — this motivates $B(c)$, it does not prove it.
- **To tabulate for review:** water flux J_w , membrane length-scale L , and diffusivity D_m per salt.

Refs — $Pe = J_w L / D_m$: DATA2, Liu et al., *Ind. Eng. Chem. Res.* 64 (2025) 12111 (eq 25); Spiegler–Kedem rejection: Spiegler & Kedem, *Desalination* 1 (1966) 311; solution-diffusion (L_p , B , σ): Kedem & Katchalsky, *Biochim. Biophys. Acta* 27 (1958) 229; framework: Ouimet et al., *J. Membr. Sci.* 641 (2022) 119743 (DATA1).

All single-salt windows sit on the Donnan plateau



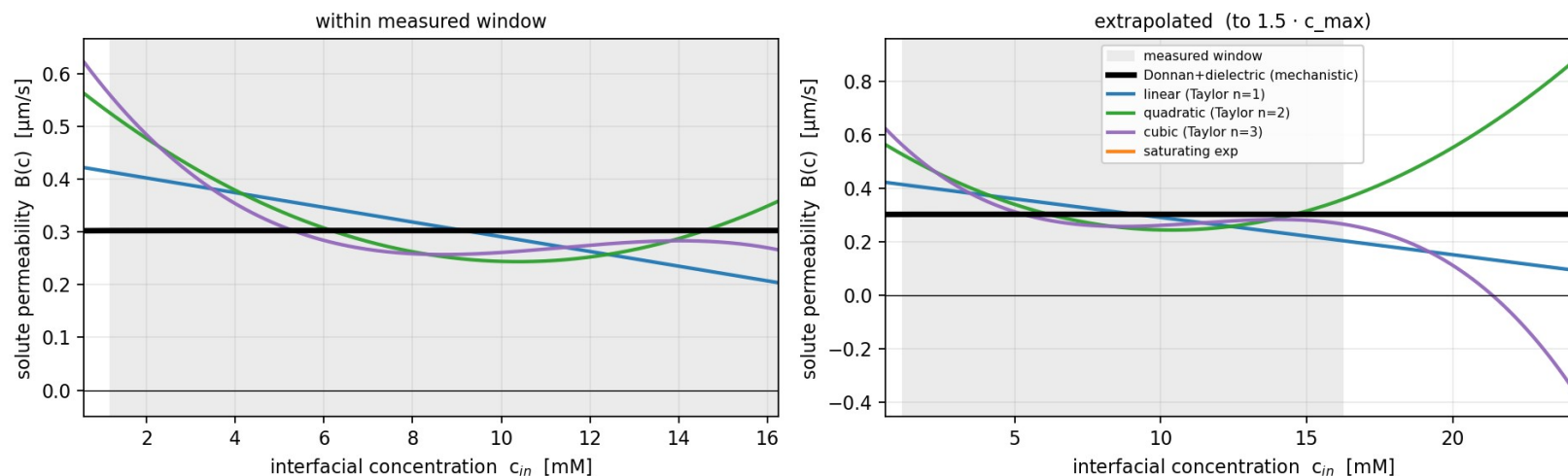
$$B(c) = J_w P_0 (-X + \sqrt{X^2 + 4k^2 c^2}) / (2c)$$

$$B_{plateau} = J_w P_0 k, \quad c_{knee} = X/2k, \quad u = c/c_{knee} = 2kc/X$$

where $B(c)$ — solute permeability at interfacial concentration c ; c — membrane-interface salt concentration; J_w — water flux; P_0 — intrinsic permeability prefactor; X — effective membrane fixed-charge density; k — lumped Donnan-dielectric partition factor; $B_{plateau}$ — high- c plateau value; $c_{knee} = X/2k$ — regime knee; $u = c/c_{knee}$ — screening parameter.

- **$B(c)$ collapses onto one master curve; every measured window is at $u \approx 10^3$ - 10^5 (screened plateau) → B is $\approx$ constant in-window (the screened-plateau value).**
- **The knee $c_{knee} = X / 2k$ rails 10^3 - $10^5 \times$ below the data → the fixed-charge scale X is not identified from one salt.**

B(c): in-window agreement, out-of-window divergence — LaCl_3



$$B(c) = \beta_0 + \beta_1 c + \beta_2 c^2 + \beta_3 c^3$$

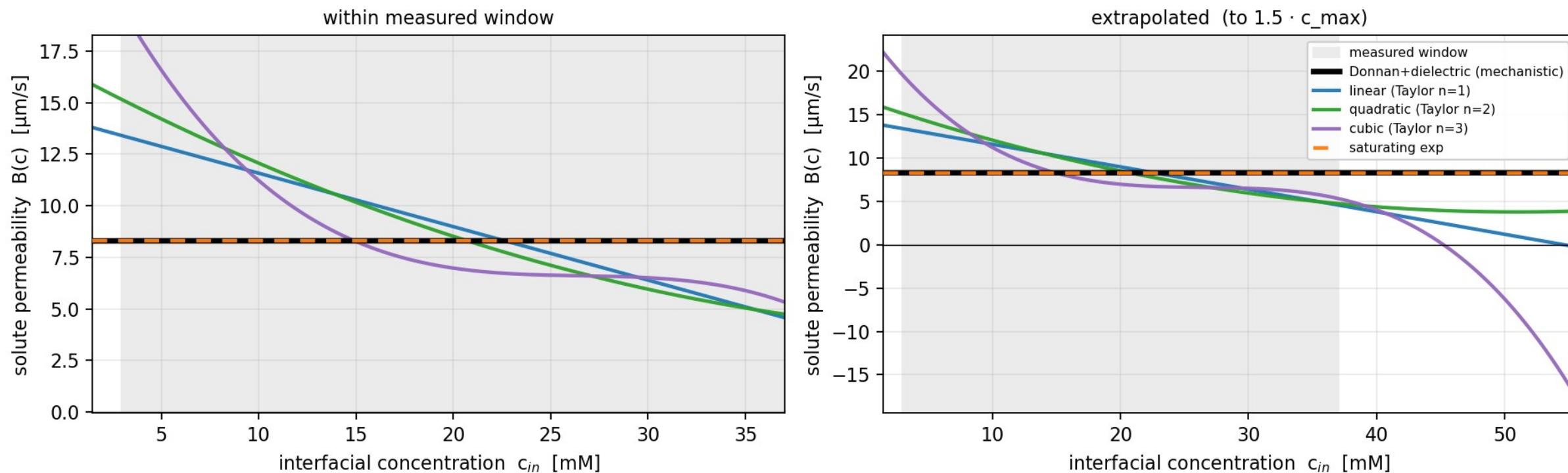
$$B(c) = B_\infty (1 - e^{-c/c^*})$$

where $B(c)$ — solute permeability; c — interfacial concentration; β_0 – β_3 — polynomial (Taylor) expansion coefficients; B_∞ — saturating permeability as $c \rightarrow \infty$; c^* — characteristic saturation concentration.

- **Within the measured window, constant / linear / quadratic / cubic / saturating-exp all agree ($B \approx 0.3 \mu\text{m}\cdot\text{s}^{-1}$).**
- **Extrapolated, the polynomials diverge unphysically (cubic $\rightarrow$ negative); Donnan and saturating-exp stay bounded.**
- **The Donnan-dielectric is the mechanistically-grounded, physically-bounded form** — the data do not discriminate it from a constant in-window.

Refs — empirical $B(c)$: polynomial (multiplicative) form per DATA2, Liu et al., Ind. Eng. Chem. Res. 64 (2025) 12111 (eq 31); saturating-exp & Donnan-bounded forms: this work (DATA3).

Same behavior for CaCl_2



- **Identical picture: in-window agreement, out-of-window polynomial divergence, Donnan bounded.**
- **On parsimony (AIC) a constant fits as well in-window** — prefer Donnan for mechanism and physical extrapolation, not in-window fit.
- **Next step to pin X:** a joint multi-salt fit (re-parameterized by ionic strength) reaches lower u and probes the knee.